Harshavardhana Gowda

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University of California, Davis

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Website

EDUCATION

Ph.D. **University of California, Davis**
Electrical and Computer Engineering
Expected: Fall 2027

B.S. **Indian Institute of Space Science and Technology**
Electrical and Computer Engineering
Fall 2018

RESEARCH EXPERIENCE

2023– **University of California, Davis**
Speech Neuroengineering Laboratory *and* Center for Mind and Brain
Graduate Student Researcher

2018–2022 **Indian Space Research Organization**
Satellite Communications and Navigation Laboratory
Research Scientist

RESEARCH AREAS

Speech neuromotor interfaces, wearable bio-interfaces.
Spoken language understanding, speech synthesis and recognition.
Human-computer interaction, understanding and creating embodied intelligence.

PUBLICATIONS

Peer Reviewed Articles

2026 **Harshavardhana T. Gowda**, Daniel C. Comstock, Lee M. Miller.
emg2speech: synthesizing speech from electromyography using self-supervised speech models.
Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers). *Project Page.*

2026 **Harshavardhana T. Gowda**, Lee M. Miller.
Non-invasive electromyographic speech neuroprosthesis: a geometric perspective.
Findings of the Association for Computational Linguistics: ACL 2026. *Project Page.*

2025 **Harshavardhana T. Gowda**, Zachary D. McNaughton, Lee M. Miller.
Geometry of orofacial neuromuscular signals: speech articulation decoding using surface electromyography.
Journal of Neural Engineering. *Project Page*.

2025 **Harshavardhana T. Gowda**, Neha Kaul, Carlos Carrasco, Marcus Battraw, Safa Amer, Saniya Kotwal, Selena Lam, Zachary D. McNaughton, Ferdous Rahimi, Sana Shehabi, Jonathon Schofield, Lee M. Miller.
A database of upper limb surface electromyogram signals from demographically diverse individuals.
Scientific Data. *Project Page*.

2024 **Harshavardhana T. Gowda**, Lee M. Miller.
Topology of surface electromyogram signals: hand gesture decoding on Riemannian manifolds.
Journal of Neural Engineering. *Project Page*.

TALKS

2026 *Non-invasive speech neuroprostheses*
University of California, Berkeley.
25th Anniversary of the CITRIS and the Banatao Institute. Theme: *"Engineering a future fit for Humanity"*.

GRANTS AND AWARDS

Competitive Grants and Fellowships

2026	*Summer Graduate Research Fellowship* at the University of California, Davis	*$22,486*
2025	*NeuralStorm Fellowship* at the University of California, Davis	*$66,551*
2024	*Department of Space Fellowship* at the Indian Institute of Space Science and Technology	*$10,000*

TEACHING

University of California, Davis

Fall 2024 Electronic Circuits and Systems. Graduate Student Instructor for 100+ students. Taught implementation of electronic circuits.

SERVICE

Academic Journal Peer Review

Journal of Neural Engineering × 3
Scientific Reports × 2
Scientific Data × 1